Exploring the Challenges of Digital Literacy Adoption in Rural Self-Help Groups: Implications for Financial Literacy and Its Impact on Financial Empowerment



P. V. Sumitha and Rowena Wright

Abstract The rapid growth of advanced technology along with the increasing demand for innovation, is affecting every sector at its core. Certain sectors, on the other hand, have adapted to and thrived with, these changes, while others have struggled or adapted poorly. Among these rural women from Self-Help-Groups (SHGs) have been left behind and face various challenges preventing them from keeping pace. This study examines how adopting digital skills among women in SHGs fosters financial inclusion by enabling active participation in banking services through financial programs, and digital payment platforms. However due to limited access to technology, low literacy levels, socio-cultural constraints, resistance to technology, and guide–at–use, the challenges are many. The paper delves into how digital fluency becomes a challenge for women in the SHGs, and the impact of the same on financial literacy and empowerment. This study surveyed 132 government-supported and 145 NGO-supported SHG women to examine levels of digital and financial literacy, adoption barriers, and the challenges of adopting digital and financial literacy. The study finds that though NGO-supported SHGs have higher digital penetration and usage due to focused training interventions, the issue with government-supported SHGs is more related to infrastructure and digital awareness. The report calls for more targeted skilling initiatives, access to devices, and localized skilling to address this disconnect. Capacity building of rural women in digital finance is an efficient means of financial empowerment in a modern & digitally savvy world and incorporating digital upskilling programs in SHGs can make an impact for progressive financial inclusion.

Keywords Digital literacy · Financial literacy · Financial empowerment

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1 Introduction

The world today is fortunately or unfortunately the consequence of the digital revolution that has created countless opportunities and some lingering problems. In developing countries, the digital divide needs to be addressed. Thus, SHGs have become potent channels for empowering women and improving their economic self-sufficiency via micro-credit and collective action. Digital literacy or the ability to use digital technology to access, evaluate, and adapt to information is critical to navigating the digital financial landscape. It aims at giving SHG women access to banking services, availing digital payment platforms, and remaining conversant with government financial programs. The understanding of financial concepts and the ability to produce sound financial decisions is extremely important. It correlates the working of digital as well as financial literacy towards financial empowerment benefitting not only with proficiency in the consumption of financial resources but also with active participation in the economy. SHG women are empowered by acquiring digital skills that enable them to make better financial decisions, access digital banking services and get easy access to financial programs. However, the penetration of digital literacy in rural regions is uneven with multiple other barriers prevalent such as socio-economic and infrastructural barriers.

The study seeks to understand the emerging challenges and barriers faced by rural SHG women to become digitally literate and its implications leading to financial literacy and financial empowerment. In this regard, the study aims to understand the challenges in adopting digital technologies and assess the role of digital training programs, inferring which would help in creating suitable reforms to diminish the digital divide and promote financial creativity. The findings of this study will help us understand how to develop practical measures to bridge the digital divide and ensure financial inclusion. Similarly, the study also toes the line with the global need for reskilling and upskilling by supporting the direct necessity of SHG women for personalized digital training to build them up, supporting financial inclusion which is crucial in the rapidly evolving digital economy.

2 Literature Review

The reviews of literature provided were structured, discussed digital literacy, financial literacy, and financial empowerment, and noted the challenges on the road to adopting digital and financial literacy towards financial empowerment.

2.1 Empowering Financial Inclusion Through Digital Literacy

Financial inclusion is wider access to financial services, which refers to the ability of individuals to save money for the future, buy insurance, and invest money, to help them achieve economic growth [5–7], this has been achieved through Digital Literacy, here through platforms that enable online banking, mobile transaction apps, and financial planning tools to ensure a multitude of options. Due to rapid technological progress in the finance field, consumers need something called digital literacy that helps them not only comprehend the financial side but also help them to defend themselves from fraud [4, 10]. Thus, getting digitally literate is a prerequisite for getting financially literate. There are several facilities for rural women who are deprived of them. One of them is a major driving factor in the implementation of digital literacy and financial literacy [20, 24] which is limited to education and opportunity. Since they have no digital skills they are deprived of financial literacy and thus they lack savings, investment, and borrowing. This can be done by promoting financial literacy in both old and new ways for fair participation in the digital financial ecosystem [4, 6]. Thus training programs for digital literacy in local languages have been designed for vulnerable populations of rural to achieve financial autonomy and participation in a digital society [15]. The financial services can be easily reached by the unbanked populations over mobile and internet banking. [9, 18]. Training of digital skills in vernacular in rural communities with support from local banks, and NGOs through SHGs has emerged as a useful tactic to tackle challenges in the adoption of digital technologies for financial inclusion. While there will be a lot of progress made to enhance digital writing in rural areas, despite the rural development there remains a high gap in digital knowledge which showcases the importance of targeted initiatives for a structured—digital literacy curriculum in the local language [8, 21]. Therefore, providing women in rural areas with financial literacy, along with digital skills, is vital to help overcome economic obstacles and encourage financial independence. Opening up the digital space for these women enables them to be active participants in the financial ecosystem, leading to better financial well-being and economic security.

2.2 Financial Literacy- Empowering Informed Choices

The developments in the digital field have privileged new opportunities for improving financial literacy and advancing financial inclusion [9]. By adopting digitalization, the policy makes financial data, functionality, and solutions more accessible, encouraging informed monetary decision-making. Thus, financial literacy is required for achieving economic well-being and financial empowerment [13]. It equips people with essential financial literacy skills regarding budgeting, saving, investing, and managing debt [1, 2]. Financial literacy is an empowering means for rural women who

have few necessary skills to make their economic contribution and to achieve gender equity [22]. By learning good financial management, people can make decisions that help them be secure financially in their life. Knowing this makes it more likely that they will save, invest, avoid debt traps, and plan for their retirement [10]. As a result, financial literacy gives them more control over their financial situation. Therefore, it plays a significant role in attaining financial security and independence [3]. Indeed, financial literacy initiatives individualized to develop pathbreaking digital literacy, specifically concerning digitized financial workspaces, have paved the way for vulnerable rural women while accounting for solid growth in their overall savings, credit(user) accessibility, and financial empowerment [12]. Strengthening the digital finance skills of rural women can help them navigate challenges with greater ease and improve their economic status.

2.3 SHG Groups: Agents of Financial Empowerment

SHGs have proved to be dynamic platforms that ensure financial accessibility and promote the sustainability of rural livelihoods. Microfinance is a key program for women's empowerment and poverty alleviation. SHGs play a significant role in achieving women's financial independence and in overall economic development [23]. SHG-Bank Linkage Program has increased savings among the poor, and the number of SHGs has increased by 45% in these years [19]. Apart from providing access to finances, SHGs also double as centers of financial literacy, providing SHG members with skills necessary for financial management [14]. SHGs have the potential to bridge the digital divide, especially for rural women and their success can be built upon. Yet, challenges like low digital and technology acceptance skills [16] and strong socio-cultural constraints on women's financial autonomy [17] prevent these skills take-up. Moreover, limited access to capital and decision-making power are additional barriers immobilizing women's entry into and engagement with digital finance. Nonetheless, SHGs can bridge the gap for digitization in the financial landscape with digital literacy training, confidence building, and behavior change interventions to be a catalyst in this process. Other non-decision augmentation issues for SHGs extend beyond just its digital adoption such as access to technology, governance issues [11], and obstacles of a different nature such as limited financial resources and the unevenness of training quality. Thus, reviews highlight that SHGs are a powerful touch point for building financial empowerment. Nonetheless, removing barriers to the adoption of digital and financial literacy is critical to reach their potential. Thus, investing in digital and financial training, and making access to digital and financial resources a way to strengthen SHGs as engines of economic progress will help rural communities benefit from the digital financial ecosystem.

3 Research Methodology

3.1 Statement of the Problem

The research builds on the significant challenges facing the women in the SHGs in rural areas in adopting digital financial literacy and the effect on financial empowerment. The present research is an attempt to prove the importance of digital literacy and financial literacy as factors contributing to financial empowerment.

3.2 Scope of the Study

The present research study is limited to rural SHG women of Dakshina Kannada district of Karnataka. It comprises both government and NGO-supported SHGs.

3.3 Research Design

The current research design is conclusive and a cross-sectional research design is adopted. Note that using a conclusive research design allows the use of both descriptive and causal research designs to determine the level of digital literacy, financial literacy, as well as financial empowerment.

3.4 Sampling Design

A multi-stage cluster sampling procedure was used to select the SHGs from the rural areas of the Dakshina Kannada district. Clusters of SHGs were identified, and from each of these clusters, a representative sample of women from both government and NGO-supported SHGs was chosen. The sample size was selected based on its practicality and the necessity to have adequate statistical power to detect significant differences between subjects. In the study, 132 women belonging to government-supported SHGs and 145 women belonging to NGO-supported SHGs were surveyed.

3.5 Research Instrument and Data Collection

A structured questionnaire was designed with closed-ended questions using standardized scales to assess digital literacy, financial literacy, and financial empowerment. There were questions for identifying the barriers to adoption and understanding

the effect of training programs as well. Qualitative data were also collected through open − ended questions. The questionnaire was administered through face-to-face interviews, which enabled the researchers to explain questions and accommodate potential literacy issues.

3.6 Field Work

Before the fieldwork, ethical clearance was provided, and informed consent was taken from every respondent. The fieldwork consisted of visiting selected SHGs and performing interviews to verify the correctness of the data. Data were collected between October 2024 and January 2025. Seva Parathinidhis was trained locally to help with data collection and translation when required.

3.7 Research Objectives

1. To measure and compare the digital literacy, financial literacy, and financial empowerment of government-supported SHGs and NGO-supported SHGs
2. To identify significant barriers to the adoption of digital and financial literacy among SHG women.
3. To analyze the challenges in the adoption of digital and financial literacy among government-supported and NGO-supported SHGs.
4. To investigate the relationship between digital literacy, financial literacy, and financial empowerment.
5. To examine the impact of digital literacy on financial literacy and financial empowerment among SHG women.

3.8 Limitations of the Study

1. Cross-sectional Study (Causality Constraints): The cross-sectional design of the study restricts the ability to derive causality relationships between variables.
2. Restricted Scope: The generalizability of findings to all rural SHG women beyond the Dakshina Kannada district may be limited.
3. Possibility of Subjectivity (Self-Reporting Bias): Dependence on self-reported data may lead to response bias.

4 Data Analysis and Interpretation

Data analysis is important in any type of research as it provides statistical evidence for the data collected. This section presents the analysis and interpretation of the data collected from the survey of 277 rural SHG women in Dakshina Kannada district.

4.1 Socio-Demographic Profile of SHG Women

The study examines various demographic variables among 277 SHG women, categorizing them into two groups: 132 SHG women from the government-supported SHGs and 145 SHG women from NGO-supported SHGs. The variables analyzed include age group, education level, marital status, family size, SHG membership duration, primary occupation, monthly household income (Rs.), access to a smartphone, access to the internet at home, prior digital training, ownership of land/assets and access to formal banking services (personal account). Table 1 below presents a comparative analysis of the demographic variables between government-supported and NGO-supported SHGs.

Interpretation: Table 1 highlights significant demographic and economic differences between government-supported and NGO-supported SHGs in the Dakshina Kannada district, focusing on economically active individuals aged 31–50.

NGO-supported SHGs predominantly consist of married members (90%), with a significant portion (46%) falling within the 41–50 age group. A notable percentage (45%) has attained secondary education, and a majority (69%) have been members for over nine years, indicating long-term engagement. These members demonstrate a shift towards self-employment (35%), resulting in slightly higher monthly incomes (above ₹10,000) and greater asset ownership. More importantly, NGO-supported SHG members have significantly better access to smartphones, home internet, and prior digital training—key factors for adopting digital financial services. In contrast, government-supported SHGs primarily engage in agriculture, with moderate education levels, larger family sizes, and relatively lower income levels. While they have good access to formal banking, they exhibit lower digital literacy rates and limited access to technology. This disparity underscores a digital divide, with government-supported SHGs facing greater challenges in accessing digital resources and training.

The socio-economic advantages observed in NGO-supported groups including higher education, self-employment, and asset ownership directly correlate with increased digital access. These findings highlight the impact of targeted NGO interventions in fostering economic empowerment and digital inclusion. They also emphasize the need for focused strategies to bridge the digital gap, ensuring government-supported SHGs gain equitable access to digital financial tools for sustained economic growth.

Table 1 Demographics of surveyed SHG women

Category	Options	SHGs	
		Government (%)	NGO (%)
Age group	18–30 years	10	8
	31–40 years	35	38
	41–50 years	43	46
	51 + years	12	8
Education level	No formal education	30	20
	Primary education	30	25
	Secondary education	35	45
	Higher secondary	5	10
Marital status	Single	2	1
	Married	85	90
	Widowed	10	7
	Divorced/ separated	3	2
Family size	1–3 members	20	25
	4–6 members	60	65
	>7 members	20	10
SHG membership duration	<3 years	5	3
	3–6 years	18	8
	6–9 years	20	20
	>9 years	57	69
Primary occupation	Agriculture/farm labor	40	30
	Self-employed (crafts, retail)	25	35
	Daily wage labor	20	15
	Homemaker	10	15
	Other	5	5
Monthly household income (Rs.)	Less than 10,000	45	18
	10,000–20,000	30	47
	>20,000	25	35
Access to smartphone	Yes	72	88
	No	28	12
Access to internet at home	Yes	15	30
	No	85	70

(continued)

Table 1 (continued)

Category	Options	SHGs	
		Government (%)	NGO (%)
Prior digital training	Yes	25	55
	No	75	45
Ownership of land/assets	Yes	30	40
	No	70	60
Access to formal banking services (Personal account)	Yes	88	97
	No	12	3

Source Primary data Analysis

4.2 Reliability and Validity Analysis

This study performed reliability and validity analysis on each of the measurement scales used in the study to make the research findings valid and reliable. Table 2 below shows the reliability analysis indicating the internal consistency of using the scales employed.

Interpretation: Cronbach's alpha is a statistic that can be calculated based on the inter-item correlation and provides an indication of how closely related a set of items. Table 2 shows Cronbach's alpha of all the factors was above better than 0.70, implying that the measurement scales of this study are reliable. In particular, digital literacy (0.85) and financial empowerment (0.88) have good internal consistency and financial literacy (0.78) have acceptable internal consistency. The results indicate that these three constructs, namely, digital literacy, financial literacy, and financial empowerment are consistent internally, and thus can outwardly be used for measuring reliability.

Interpretation: Exploratory Factor Analysis (EFA) with varimax rotation was performed to assess construct validity. The results shown in Table 3 indicate strong construct validity for all three of the scales as items loaded highly on their factors at a

Table 2 Reliability analysis

Construct	Number of items	Cronbach's alpha	Interpretation
Digital literacy	10	0.85	Good internal consistency
Financial literacy	10	0.78	Acceptable internal consistency
Financial empowerment	10	0.88	Good internal consistency

Source Primary Data Analysis

Table 3 Convergent and discriminant validity analysis (correlation matrix)

Construct	Digital literacy	Financial Literacy	Financial Empowerment
Digital literacy	1.00	0.65**	0.58**
Financial literacy	0.65**	1.00	0.72**
Financial empowerment	0.58**	0.72**	1.00

Source Primary Data Analysis
Note ** $p < 0.01$

loading greater than 0.70. The adjusted variance for each factor, which indicates the respective score obtained, ranged from 60 to 70, demonstrating that the scales reliably measure the target constructs. This demonstrates that the scales measure digital literacy, financial literacy, and financial empowerment. Pearson's correlation matrix shows that all three variables have a strong positive correlation with each other. In particular, digital literacy and financial literacy ($r = 0.65$), digital literacy and financial empowerment ($r = 0.58$), and financial literacy and financial empowerment ($r = 0.72$) are statistically significant ($p < 0.01$). These results suggest convergent validity, where constructs that should be related are correlated. Since the correlations are below 0.90, we further conclude that the constructs are distinct enough to support discriminant validity.

4.3 Assessment of Digital Literacy, Financial Literacy, and Financial Empowerment Levels Among Two Groups of SHGs

To assess digital literacy, financial literacy, and financial empowerment levels among two groups of SHGs—government-supported and NGO-supported—descriptive statistics were used. Additionally, independent sample t-tests were conducted to test the following hypotheses. Table 4 presents descriptive statistics for digital literacy, financial literacy, and financial empowerment levels among the two groups of SHGs, while Table 5 displays the results of the independent samples t-test.

H1*: There is a statistically significant difference in mean digital literacy scores between government-supported and NGO-supported SHGs.*

H2*: There is a statistically significant difference in mean financial literacy scores between government-supported and NGO-supported SHGs.*

H3*: There is a statistically significant difference in mean financial empowerment scores between government-supported and NGO-supported SHGs.*

Interpretation: Table 4 demonstrates a consistent pattern: NGO-supported SHGs exhibit higher mean scores across all three measured variables. Specifically, they demonstrate higher mean scores for digital literacy (4.20 vs. 3.50), financial literacy

Table 4 Descriptive statistics for digital literacy, financial literacy, and financial empowerment

Variable	SHG	N	Mean	SD
Digital literacy	Government	132	3.50	1.20
	NGO	145	4.20	1.00
Financial literacy	Government	132	3.00	1.10
	NGO	145	3.80	1.00
Financial empowerment	Government	132	2.80	1.00
	NGO	145	3.50	0.90

Source Primary Data Analysis

Table 5 Independent samples t-test results for digital and financial literacy

Variable	SHG	N	Mean	SD	Levene's test (F, p)	t (df)	p	Cohen's d
Digital literacy	Government	132	3.50	1.20	1.20, 0.274	−4.56 (275)	<0.001	−0.48
	NGO	145	4.20	1.00				
Financial literacy	Government	132	3.00	1.10	0.85, 0.357	−5.23 (275)	<0.001	−0.55
	NGO	145	3.80	1.00				
Financial empowerment	Government	132	2.80	1	0.90,0.344	−5.00 (275)	<0.001	−0.53
	NGO	145	3.50	0.90				

Source Primary Data Analysis

(3.80 vs. 3.00), and financial empowerment (3.50 vs. 2.80) when compared to government-supported SHGs. These findings suggest that NGO-supported SHGs possess greater knowledge, skills, and capabilities required to navigate the digital financial landscape and make informed financial decisions. Furthermore, the higher minimum scores observed in NGO-supported SHGs indicate a higher level of competence in these areas. These descriptive statistics support the conclusion that NGO interventions most likely play a significant role in enhancing digital and financial capabilities among rural SHG women.

Interpretation:

Digital Literacy: Levene's test for equality of variances was not significant ($F = 1.20$, $p = 0.274$), indicating that the variances of digital literacy scores were statistically similar between the government-supported and NGO-supported SHGs. Consequently, the "equal variances assumed" row of the t-test output was used. The independent samples t-test revealed a statistically significant difference in digital literacy scores between the two groups ($t(275) = -4.56$, $p < 0.001$). Specifically, the mean digital literacy score was significantly higher for NGO-supported SHGs (Mean = 4.20, SD = 1.00) compared to government-supported SHGs (Mean = 3.50, SD = 1.20). Cohen's $d = -0.48$ indicates a medium effect size, signifying a practically meaningful difference. Therefore, we reject the H1 hypothesis and

conclude that NGO-supported SHGs exhibit significantly higher digital literacy than government-supported SHGs.

Financial Literacy: Similarly, Levene's test was not significant ($F = 0.85$, $p = 0.357$), confirming equal variances for financial literacy scores. The independent samples t-test demonstrated a statistically significant difference in financial literacy scores between the two groups ($t(275) = -5.23$, $p < 0.001$). The mean financial literacy score was significantly higher for NGO-supported SHGs (Mean = 3.80, SD = 1.00) compared to government-supported SHGs (Mean = 3.00, SD = 1.10). Cohen's d of −0.55 indicates a medium effect size, implying a practically meaningful difference. Thus, we reject the H2 hypothesis and conclude that NGO-supported SHGs possess significantly higher financial literacy than government-supported SHGs.

Financial Empowerment: Levene's test for equality of variances was not significant ($F = 0.90$, $p = 0.344$), indicating that the variances of financial empowerment scores were not significantly different between the two groups. Therefore, the "equal variances assumed" row of the t-test output was used. The independent samples t-test revealed a statistically significant difference in financial empowerment scores between government-supported and NGO-supported SHGs ($t(275) = -5.00$, $p < 0.001$). The mean financial empowerment score was significantly higher for NGO-supported SHGs (Mean = 3.50, SD = 0.90) compared to government-supported SHGs (Mean = 2.80, SD = 1.00). Cohen's d of −0.53 indicates a medium effect size, suggesting a practically meaningful difference. Thus, we reject the null hypothesis and conclude that NGO-supported SHGs exhibit significantly higher financial empowerment compared to government-supported SHGs.

The results of the independent samples t-tests consistently demonstrate that NGO-supported SHGs exhibit significantly higher levels of digital literacy, financial literacy, and financial empowerment compared to government-supported SHGs. These findings suggest that NGO interventions are likely more effective in enhancing digital literacy, financial literacy, and financial empowerment among SHG members. The medium Cohen's d values for all three variables indicate that these observed differences are not only statistically significant but also practically meaningful, highlighting the real-world impact of the disparities.

4.4 Adoption Barriers to Digital Literacy and Financial Literacy

Table 6 presents the key barriers to digital literacy adoption among government-supported and NGO-supported SHGs, highlighting significant disparities between the two groups. Table 7 presents the key barriers to financial literacy adoption among government-supported and NGO-supported SHGs, highlighting the stark differences between the two groups.

Table 6 Adoption barriers to digital literacy

Adoption barriers to digital literacy	Government-supported (%)	NGO-supported (%)
Limited access to smartphones or digital devices	65	30
Lack of internet connectivity	70	40
Low digital literacy and lack of technical skills	80	50
Fear of technology and lack of confidence in using digital tools	55	35
Language barriers due to lack of local language support	40	20
Socio-cultural restrictions limiting digital engagement (e.g., family norms)	35	25
Inadequate access to relevant digital literacy training programs	75	45

Source Primary data analysis

Table 7 Adoption barriers to financial literacy

Adoption barriers to financial literacy	Government supported (%)	NGO-supported (%)
Lack of understanding of financial concepts	75	40
Restricted access to financial information	60	30
Insufficient practical experience in financial management	80	55
Fear of financial risks/losses	50	30
Low literacy and numeracy skills	65	35
Inadequate financial training programs	70	40
Socio-cultural restrictions on financial decision-making	45	25

Source Primary data analysis

Interpretation: Table 6 reveals substantial differences in digital literacy adoption barriers between government-supported and NGO-supported SHGs. Government-supported SHGs face greater challenges, with the most significant obstacles being: a lack of digital literacy and lack of technical skills (80%), limited access to smartphones or digital devices (70%), and inadequate access to relevant digital literacy training programs (75%). In contrast, NGO-supported SHGs report lower barriers across all categories, indicating better resource allocation, targeted interventions, and localized support.

To overcome digital literacy adoption barriers, SHGs should prioritize comprehensive digital training programs, especially in local languages, address socio-cultural constraints, including family restrictions that limit women's access to digital tools,

and boost confidence-building initiatives to help members overcome fear of technology and enhance digital participation. By implementing these strategies, SHGs can bridge the digital divide among rural communities.

Interpretation: Among the barriers to financial literacy adoption, Table 7 shows significant differences between government-supported and NGO-supported SHGs. Government-supported SHGs encounter greater hurdles, including insufficient practical financial management experience (80%), limited comprehension of financial concepts (75%), and lack of proper financial training programs (70%). Compared with NGO-supported SHGs, they report much higher barriers on all dimensions, providing further evidence that NGO interventions (such as better financial training, access to financial information, and opportunities for hands-on learning) result in lower barriers to financial literacy. Further challenges of fear of financial risks, limited skills of literacy, and socio-cultural barriers are also pervasive with government-supported SHGs, which could be attributed to less customized financial education and support mechanisms. Comparing barriers to digital and financial literacy, the report finds that compared to NGO-supported SHGs, government-supported SHGs face more challenges in implementing both digital and financial literacy initiatives. Government-supported SHGs can address this gap by scaling targeted and personalized interventions to instill the requisite level of digital and financial literacy among the members, allowing them to participate more in the economy and also improving the financial inclusion level.

4.5 Challenges in Adoption of Digital and Financial Literacy Among SHG Groups

Table 8 presents the key challenges in the adoption of digital and financial literacy among government-supported and NGO-supported Self-Help Groups (SHGs), illustrating significant differences in access, skills, and socio-cultural constraints.

Interpretation: The findings from above Table 8 reveal significant disparities between government-supported and NGO-supported SHGs regarding the challenges faced in adopting digital and financial literacy. Findings infer that government-supported SHGs report higher levels of difficulty across all identified challenges, emphasizing the need for targeted interventions to enhance their digital and financial capabilities. The results underscore the importance of structured training programs, hands-on experience, and confidence-building initiatives in overcoming digital and financial literacy challenges. To bridge the gap, government-supported SHGs should focus on: implementing well-structured, high-quality digital and financial training programs, ensuring that SHG members have access to smartphones, stable internet, and user-friendly digital financial platforms, conducting awareness campaigns and mentorship programs to help members feel more comfortable with digital and financial services and strengthening hands-on learning in digital transactions, budgeting exercises, and

Table 8 Challenges in the adoption of digital and financial literacy among SHG groups

Category	Challenge	Government SHGs (%)	NGO SHGs (%)
Digital literacy	Low confidence and fear of technology	65	30
	Limited hands-on experience with digital financial services	70	40
	Security concerns and fear of fraud	80	50
	Inconsistent or low-quality training programs	75	35
	Resistance to change and adaptation	40	20
Financial literacy	Difficulty in applying financial knowledge to real-life situations	75	40
	Fear of financial risks and making wrong decisions	60	30
	Lack of hands-on financial experience	80	55
	Low confidence in handling financial transactions	50	30
	Limited guidance and inadequate financial training	65	35

Source Primary Data Analysis

small-scale investments. By implementing these strategies, government-supported SHGs can improve digital and financial literacy levels, ultimately fostering economic empowerment and financial inclusion for rural women.

4.6 Correlation and Regression Analysis

The correlation analysis examines the relationship between digital literacy, financial literacy, and financial empowerment among SHG women. Whereas regression analysis is used to predict digital literacy and financial literacy are the significant predictors of financial empowerment.

H4: Increase in digital literacy increases financial literacy among SHG women.

H5: Increase in digital literacy increases financial empowerment among SHG women.

H6: Increase in financial literacy increases and financial empowerment among SHG women.

H7: Digital literacy and financial literacy are significant predictors of financial empowerment.

Interpretation: Table 9 presents the correlation matrix for digital literacy, financial literacy, and financial empowerment, highlighting significant positive relationships between these variables. Results highlight the moderately strong positive correlation between digital literacy and financial literacy ($r = 0.65$) suggesting higher level of literacy on cash-less and digital currency might increase financial literacy. This means that women with better digital skills are more likely to understand financial concepts and participate in financial activities. In the same vein, the relationship between digital literacy and financial empowerment ($r = 0.58$) suggests a moderate positive correlation, signifying the empowering nature of digital education on financial capability. The correlations of financial literacy and financial empowerment ($r = 0.72$) suggest that digital literacy training should be included in financial education to enhance financial empowerment. Though there is a positive relation between financial literacy and financial empowerment, interventions can focus on financial education and the use of digital tools that make it more accessible and engaging. The measure of success for rural SHG women on the road from digital literacy to digital equity and from financial inclusion to financial empowerment lies in their ability to generate digital footprints through active participation in financial ecosystems.

Interpretation: Table 10 highlights the R-value of 0.74 which indicates a strong positive correlation between the predictors, digital literacy and financial literacy, and the outcome variable, financial empowerment. The R-squared value of 0.55 signifies that 55% of the variance in financial empowerment is explained by the model, demonstrating that these predictors are important determinants of financial empowerment. The highly significant F-statistic ($F(2, 274) = 168.00$, $p < 0.001$) confirms that the overall regression model is statistically significant, indicating a good fit for the data. Both digital literacy ($t = 6.00$, $p < 0.001$) and financial literacy ($t = 7.50$, $p < 0.001$) exhibit highly significant t-statistics, confirming their strong predictive power

Table 9 Correlation matrix for digital literacy, financial literacy, and financial empowerment

Construct	Digital literacy	Financial literacy	Financial empowerment
Digital literacy	1.00		
Financial literacy	0.65**	1.00	
Financial empowerment	0.58**	0.72**	1.00

Source Primary Data Analysis
Note ** $p < 0.01$

Table 10 Regression analysis predicting financial empowerment

Predictor	B (Unstandardized coefficient)	SE B (Standard error)	B (Standardized coefficient)	T	P
Constant	0.50	0.20	–	2.50	0.013
Digital literacy	0.30	0.05	0.35	6.00	<0.001
Financial literacy	0.45	0.06	0.50	7.50	<0.001

Dependent variable: Financial empowerment. $R = 0.74$; $R2 = 0.55$; $F(2, 274) = 168.0$, $p < 0.001$.
Source Primary Data Analysis.

for financial empowerment. Comparing the standardized coefficients (β), financial literacy ($\beta = 0.50$) demonstrates a slightly stronger predictive power than digital literacy ($\beta = 0.35$) in explaining financial empowerment. Consequently, the regression analysis confirms that both digital literacy and financial literacy are significant positive predictors of financial empowerment among the surveyed SHG women, with the model explaining a substantial 55% of the variance. Financial literacy emerges as having a slightly stronger influence. These findings underscore the critical importance of promoting both digital and financial literacy skills to enhance financial empowerment among rural women.

5 Discussions

The present study proved there are significant barriers to the adoption of digital literacy and financial literacy among SHG women, NGO-supported programs have successfully aided in these women's digital and financial education, and the success of these programs can largely be attributed to their personalization. The adoption of digital and financial literacy is particularly undermined by limited internet connectivity, lack of comparative financial and digital training programs, and limited practical experience in financial management. To fill this gap, there is a need for organized financial and digital training courses in regional languages for SHG women. The results from the study demonstrate the challenges of adoption of digital and financial literacy remain pronounced as SHGs supported by the government and NGOs differ immensely in terms of SHG performance. SHGs with less government backing tend to face more significant challenges, such as low confidence in the use of technology, minimal hands-on experience, security concerns, and fragmented training programs. These hurdles limit their potential for meaningful interaction with digital financial services or informed choices with their money. NGO-supported SHGs show higher resilience and adaptability characteristics as they receive structured training and mentorship programs, and have better access to digital tools. SHGs supported by the government face limited financial independence due to barriers to financial literacy, including challenges in implementing knowledge or fear of financial risks.

To eliminate these hurdles, the pathway must be practical training, mentorship, and resources. Digital and financial literacy have a role in contributing to economic empowerment and promoting rural financial inclusion.

6 Conclusion

This study highlights how basic digital literacy can empower rural SHG women by promoting financial inclusion. However, significant disparities in digital adoption persist, with NGO-led programs demonstrating greater effectiveness due to their tailored approaches. Addressing this digital divide will require urgent action, with a structured approach. This entails overcoming major obstacles such as limited internet connectivity, costly devices, as well as the demand for localized training and language assistance.

The findings underline the need for collaborative action by the government, financial institutions, and NGOs to develop specific digital interventions that boost financial capabilities. Sustainable digital financial empowerment depends on tailored training, available tech infrastructure, and ongoing support systems. When it comes to empowering rural women financially and professionally, transforming their minds to learn digital finance not only helps the business grow but also allows the women to be competitive in their industry. This demonstrates the importance of embedding digital upskilling initiatives as a core part of SHGs, to allow rural women to bridge the skills gap while contributing to broader goals of sustainable financial inclusion and following the same journey that male members of their families do in the digital economy.

Ultimately, digital literacy fosters financial literacy, which creates a globally competitive and financially inclusive society. The long-term effects of these interventions and more research on the areas of mobile technology to drive the financial inclusion of rural women should be part of future research agendas.

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